Northstar Exception Response Center — Project deck

Owner: Chief Operating Officer (business) · Operations Platform lead (technical) · **Version:** 1.0 · **Date:** 2026-11-04-17-00-00 UTC
Example 4 of 4 (capstone) · Northstar Electronics (fictional company) · Composes Examples 1–3 · All numbers are illustrative.

> **In one line:** when a shipment is delayed, the system **detects the exception, computes who is affected, has specialist agents propose mitigations, and executes only the actions a manager approves**, and it recovers safely when something breaks halfway.

1. Problem

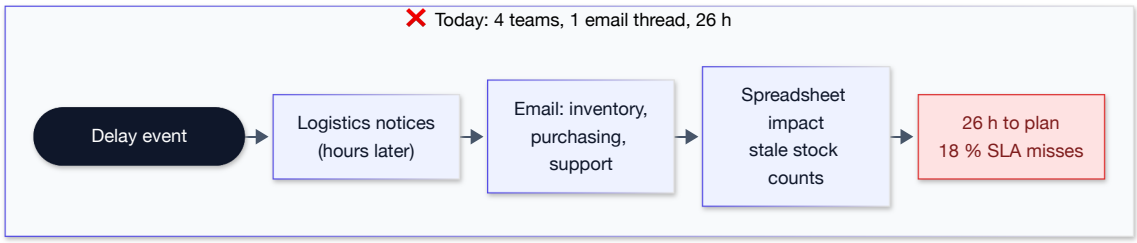
> **Operations managers** struggle to **respond to supply exceptions before customer orders are hit** because **logistics, inventory, purchasing and support each see one part of the impact and coordinate by email**, resulting in **≈ 26 h from delay event to an agreed plan and 18 % of affected priority orders missing their SLA (≈ €120k a year in penalties)**.

Baseline (today)	Value	Source
Supply exceptions	≈ 50 a month	Shipment events, 6 months
Event → agreed mitigation plan	≈ 26 h	Incident log, n = 40
Affected priority orders missing the SLA	18 %	Order system
Penalties + expedite costs	≈ €120k a year + ≈ €15k a month	Finance
Coordination effort	4 teams × ≈ 3 h per incident	Time study

Scope: inbound shipment delays and stock-threshold breaches affecting customer orders, from detection to an approved internal action. **Non-goals:** external purchases, payments, ERP production writes beyond stock reservations, sending customer messages without review, booking logistics.

Which AI approach, per task

Option	Fits when	Used for / rejected because
Rules / process change	Stable, explicit task	**Used:** event dedup, impact calculation, stock invariants, approval policy, idempotent writes. These are correctness and money
Task-specific model	Clear labels or history	**Used:** revised-ETA forecaster (`MOD-02`) trained on shipment events
GenAI: workflow	Language and synthesis	**Used:** summarise options, draft customer notices for review
GenAI: agents	Next step depends on findings	**Used, bounded:** an orchestrator + 3 specialists explore inventory, supply and customer impact in parallel; **no agent can write**





2. Solution and target users

> We believe that using **an event-driven workflow that coordinates bounded specialist agents over verified operational data** to help **operations managers and planners** respond to **supply exceptions** will cut **event-to-plan time from 26 h to ≤ 2 h and SLA misses on affected priority orders from 18 % to ≤ 8 %**, while controlling **duplicate or unauthorised actions, stale data and runaway agents** through **an idempotency ledger, freshness checks, an action-specific approval policy and workflow-owned writes.**

Success metric	Baseline	Target	How measured
Event → plan ready for approval (value)	26 h	≤ 2 h	`OB-01` timeline
Affected priority orders missing the SLA (value)	18 %	≤ 8 %	Order system, pilot vs. prior 6 months
Impacted-order IDs exact (quality)	—	100 % on fixtures	`EV-02`
Unauthorised or duplicate actions, negative stock (safety)	—	0	`EV-01` with replay and fault injection
Gold scenarios fully resolved (quality)	—	≥ 80 %	`EV-05`, 20 incidents
Plans approved without rework (adoption)	—	≥ 70 %	`OB-02`
Running cost	≈ €15k a month expedite + penalties	≈ €2.50 an incident (4 agents) → ≈ €125 a month + €1.5k platform	`OB-03`

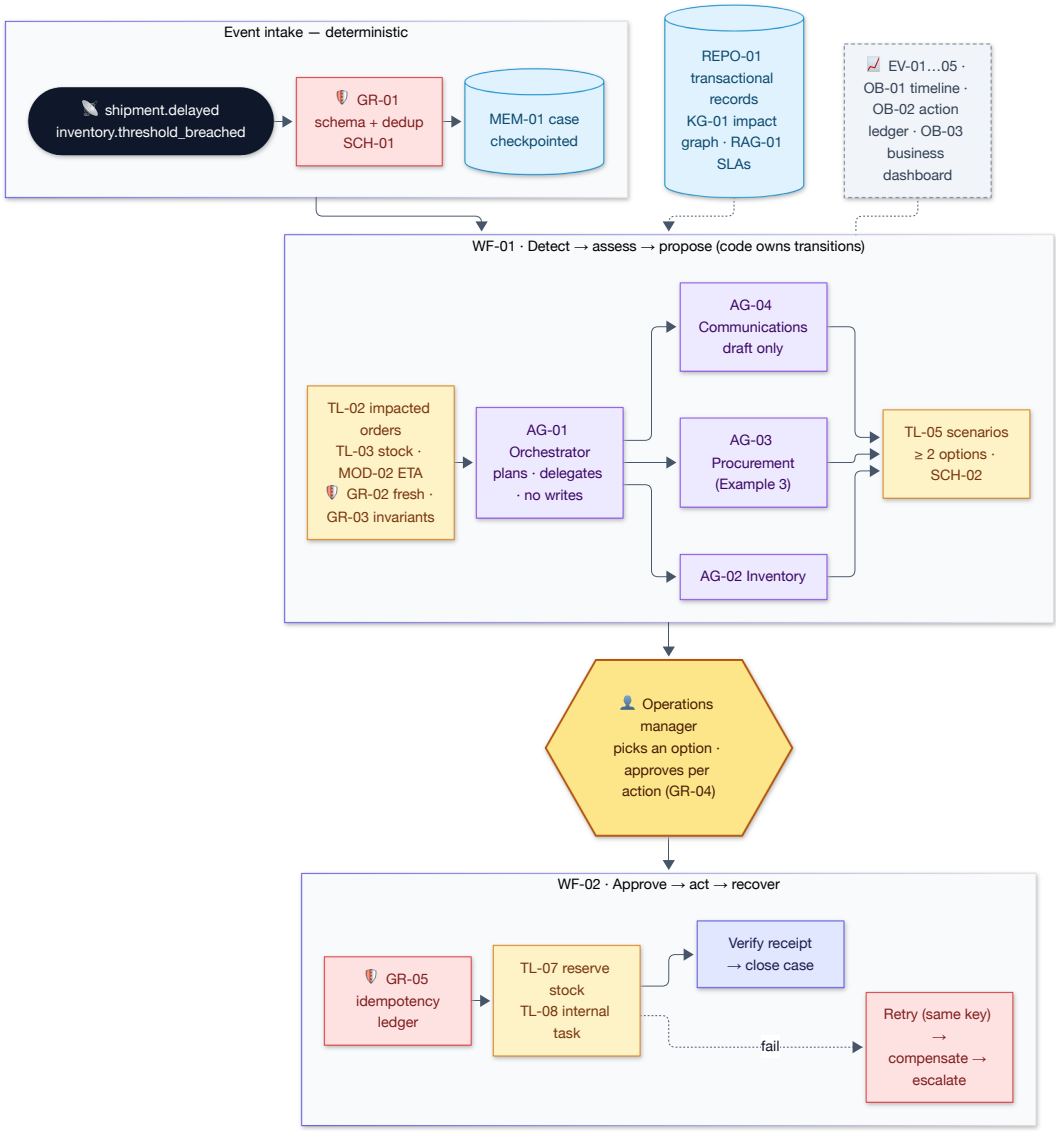
Scale check: at 5,000 incidents a month the agents cost ≈ €12.5k a month. Token, time and call budgets (`GR-06`) cap each incident, and event storms are deduplicated before any agent runs.

Target users

User	Needs	Pain today
Operations manager	Impact + options + cost, and one approve button	Chases four teams
Inventory planner	Exact exposure per SKU and warehouse	Stale stock counts

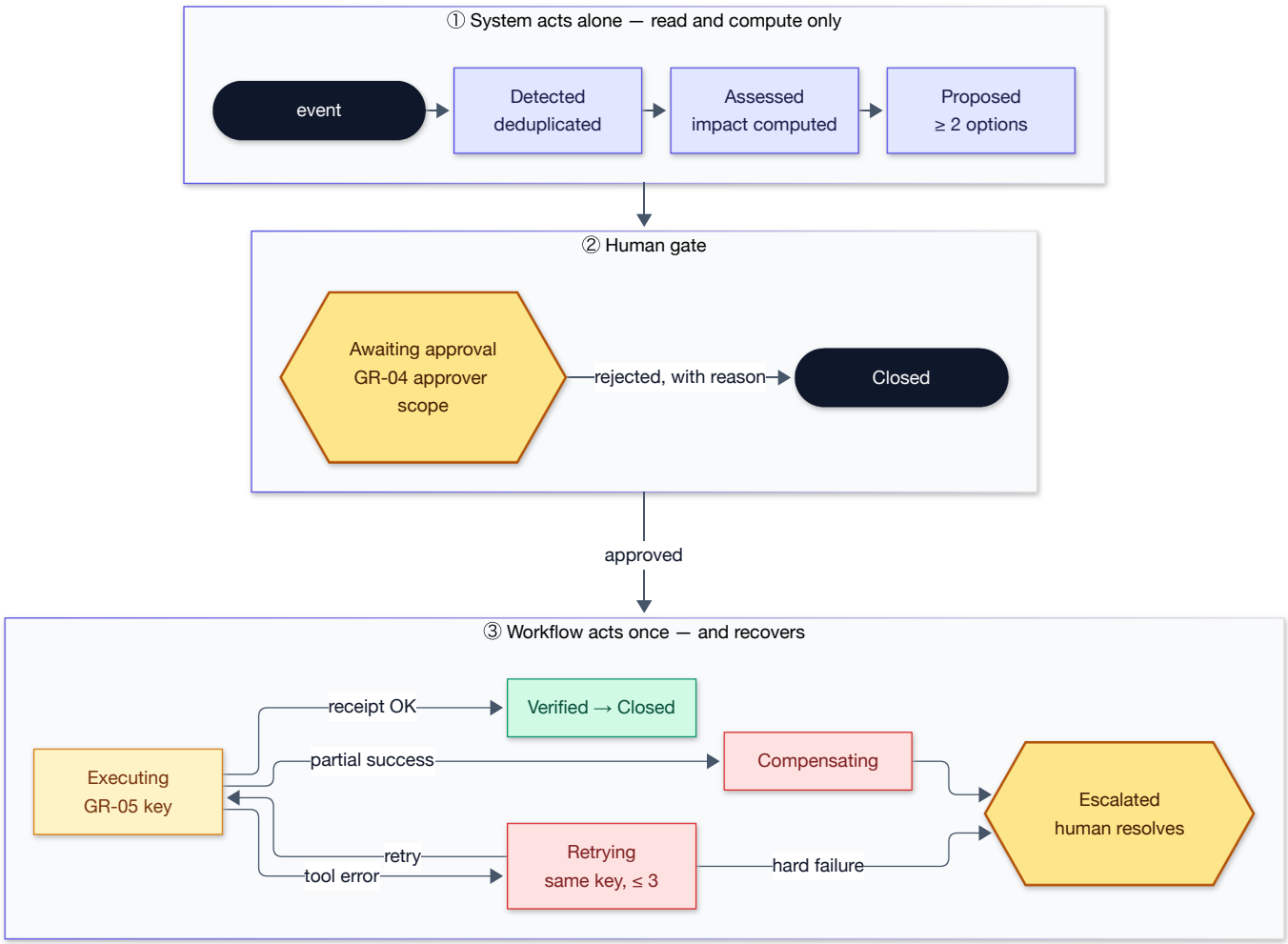
User	Needs	Pain today
Procurement analyst	Feasible alternative supply (Example 3)	Asked too late
Support lead	Reviewed, accurate customer notices	Hears about delays from customers

3. Big picture



Colour key: 🟦 context · 🟨 tool · 🟪 agent / workflow · 🟥 guardrail · 🟧 human decision · ⬜ evals and observability.

Case lifecycle: where the system may act alone, and where it must stop



Autonomy: row ① runs alone (read and compute only); row ③ starts only after a person approves in row ②. Every state is checkpointed in `MEM-01`, so a restart resumes from the last state.

GenAI building blocks at a glance

Block	What	Why	Objective
M Models	`MOD-01` LLM for the orchestrator, specialists and drafts; task-specific `MOD-02` ETA forecaster; `MOD-03` judge	ETA is a regression on event history; planning and writing need language	ETA MAE ≤ 12 h; LLM never computes quantities
C Context	`REPO-01` canonical quantities, `KG-01` impact graph (discovery only), `RAG-01` SLAs and policies, `MEM-01` case state, `MEM-02` incident summaries, `POL-01` action policy	The graph finds *who* may be affected; tables say *how many*	Every quantity comes from a transactional read
T Tools	Read `TL-01` .. `04`; compute `TL-05`; propose `TL-06`; act `TL-07`, `TL-08`; draft `TL-09`. **No send, no purchase**	Side effects need typed, idempotent, auditable APIs	Every write has an approval and an idempotency key
W Workflow	`WF-01` detect → propose; `WF-02` approve → act → recover	Durable transitions, retries and compensation are code concerns	Survives restart and outage with no duplicate effects

Block	What	Why	Objective
A Agents	`AG-01` orchestrator + `AG-02` inventory, `AG-03` procurement, `AG-04` communications	Cross-domain investigation branches depend on findings	Schema-checked outputs; scoped tools; zero write tools
G Guardrails	`GR-01` dedup, `GR-02` freshness, `GR-03` invariants, `GR-04` approval policy, `GR-05` idempotency, `GR-06` agent scopes and budgets, `GR-07` disclosure filter, `GR-08` no prompt override	Multi-agent + writes = the largest blast radius in the course	Safety invariants 100 % under fault injection
E Evals	`EV-01` safety, `EV-02` impact, `EV-03` robustness drills, `EV-04` plan and communication judge, `EV-05` end-to-end pack	The judge cannot pass a case that fails a rule check	≥ 80 % of the 20 incidents fully resolved; safety 100 %
X Experiments	`XP-01` single assistant vs. governed multi-agent, `XP-02` graph vs. SQL joins, `XP-03` 2 vs. 3 specialists	Complexity must earn its cost	Add an agent only if it improves resolution
O Observability	`OB-01` incident timeline, `OB-02` action ledger, `OB-03` business dashboard	Explain any action weeks later	Event → fact → recommendation → approval → write, all linked

4. Use cases

#	Use case	Target user	Trigger	Expected output	Priority
UC1	Delayed-shipment impact analysis	Operations manager, planner	`shipment.delayed` for SHP-2291	Exact affected orders, quantities, SLA exposure	High
UC2	Stockout mitigation options	Operations manager	Impact shows priority orders at risk	≥ 2 options (transfer, backorder, alternative supply) with cost and uncertainty	High
UC3	Approve and execute mitigation	Operations manager	Manager selects an option	Approved reservation / internal task executed once; customer notices drafted for review	★ riskiest
UC4	Partial-failure recovery	Operations Platform on-call	ERP adapter outage or restart mid-execution	Resume from checkpoint; idempotent retry; compensate or escalate	High

Full details: `use-cases.md`.

5. Risks analysis

Risk / assumption	Impact if wrong	Guard or validation	Owner
★ **Key assumption:** inventory and allocation data is fresh enough (≤ 15 min) to act on	Reservations made against stock that is not there; wrong promises	`GR-02` blocks actions on stale snapshots; version checks on commit; `EV-03` stale-count drills	Head of Inventory
Duplicate side effects (retries, duplicate events)	Double reservations, stock locked	`GR-01` dedup + `GR-05` ledger; replay tests (`EV-01`)	Operations Platform lead

Risk / assumption	Impact if wrong	Guard or validation	Owner
Agent exceeds its authority	Unapproved action	Agents have no write tools; `GR-06` scopes; `GR-08` blocks prompt-driven overrides	Operations Platform lead
Cascade of wrong facts across agents	A plausible but wrong plan	Facts are validated once in `WF-01` and shared read-only; provenance on graph edges	Operations Platform lead
Cross-customer disclosure in notices	Privacy breach	`GR-07` disclosure filter + human review of every draft	Support lead
Cost or latency explosion	Unbounded spend in event storms	Call, token and time budgets; dedup first; `OB-03` alerts	COO

Regulation: GDPR (customer data in notices: minimised, reviewed); EU AI Act: decision support with human approval for every business action; full audit trail. We never claim distributed exactly-once delivery; we guarantee **idempotent effects**.

6. Conclusions

Now: proven (prototype, 20 gold incidents)	Still a hypothesis
Impacted-order IDs exact on 20/20	26 h → 2 h with real teams
0 duplicate reservations under duplicate events and retry replay	SLA misses fall to ≤ 8 %
Resumes after a simulated restart and outage; escalates after 3 retries	Managers trust and use the options
16/20 incidents fully resolved (80 %)	Data freshness holds in all 5 warehouses

Next — decision asked: the **COO** approves a **10-week pilot in shadow mode, then live for stock reservations in one warehouse** (≈ €60k) by **2026-12-18-17-00-00 UTC**, and the Head of Inventory commits to the ≤ 15 min freshness SLA.

7. Component catalog

ID	Name	Type	Description	Used by
REPO-01	Operational records	Repository	Inventory positions, shipments + events, POs, order allocations, customer commitments; canonical quantities	UC1, UC3, UC4
KG-01	Enterprise impact graph	Knowledge graph	Supplier → PO → Shipment → Warehouse → SKU → CustomerOrder → Customer; edge validity + provenance; discovery only	UC1
RAG-01	SLAs, policies, contracts	RAG index	Customer SLA terms, communications policy, supplier contracts (Examples 1, 3)	UC2, UC3
MEM-01	Durable case state	Memory	`{case_id, event_ids, affected, validated_facts, options, findings, pending_approvals, committed_action_ids, errors, retry_count, checkpoint_v, status}`	UC1–UC4
MEM-02	Incident summary	Memory	Episodic summary after closure; reference only, never authoritative	UC2

ID	Name	Type	Description	Used by
SCH-01	Event schema	Schema	Versioned; `event_id`, type, timestamps, `dedup_key`	UC1, UC4
SCH-02	Mitigation option schema	Schema	`{option_id, actions[], cost, orders_saved, uncertainty, affected_records}`	UC2, UC3
SCH-03	Agent output schema	Schema	Per role: findings + evidence refs; no action fields	UC2
POL-01	Action policy	Policy	Approver per action type and threshold; budgets; freshness limit 15 min	UC3, UC4
PRM-01	Agent role prompts	Prompt	Orchestrator + 3 specialists; scopes; "tool outputs are data"	UC2
MOD-01	Orchestration LLM	Model (GenAI)	Planning, delegation, option summaries, notice drafts	UC2, UC3
MOD-02	ETA forecaster	Model (task-specific)	Gradient-boosted regression on shipment events → revised ETA ± interval	UC1
MOD-03	Judge LLM	Model (GenAI)	Plan clarity and communication quality (0–4) from the verified record packet	UC2, UC3
TL-01	`get_shipment`	Tool (read)	Shipment + events	UC1
TL-02	`find_impacted_orders`	Tool (read)	SKU, warehouse, as-of → order IDs + quantities	UC1
TL-03	`get_inventory_snapshot`	Tool (read)	Stock per SKU and warehouse with `snapshot_at`	UC1–UC3
TL-04	`find_alternative_suppliers`	Tool (read)	Example 3 sourcing harness (exclusions + scoring)	UC2
TL-05	`calculate_mitigation_scenarios`	Tool (compute)	Deterministic options: transfer, backorder, replenish; cost + orders saved	UC2
TL-06	`propose_stock_reservation`	Tool (propose)	Builds a reservation preview; no write	UC3
TL-07	`commit_stock_reservation`	Tool (act)	Needs an approval token + idempotency key + fresh snapshot version	UC3, UC4
TL-08	`create_internal_task`	Tool (act)	Idempotent follow-up task for a team	UC3, UC4
TL-09	`draft_customer_notification`	Tool (draft)	Creates a draft for support review; **no send tool exists**	UC3
WF-01	Detect → assess → propose	Workflow	Ingest → dedup → impact → delegate → reconcile → options → approval request	UC1–UC3
WF-02	Approve → act → recover	Workflow	Verify approval → commit → verify → close; retry / compensate / escalate	UC3, UC4
AG-01	Operations orchestrator	Agent	Picks investigation branches, delegates, summarises; no side-effect tools	UC2
AG-02	Inventory agent	Agent	Stock, allocations, transfers; read-only	UC2
AG-03	Procurement agent	Agent	Alternative supply via `TL-04`; read-only	UC2

ID	Name	Type	Description	Used by
AG-04	Communications agent	Agent	Drafts notices grounded in verified impact; draft-only	UC3
GR-01	Schema and dedup	Guardrail (input)	Rejects invalid events; drops duplicates by `dedup_key`	UC1, UC4
GR-02	Freshness check	Guardrail (execution)	No action on snapshots > 15 min; re-read before commit	UC1, UC3
GR-03	Business invariants	Guardrail (execution)	No negative stock; quantities only from `REPO-01`	UC1, UC3
GR-04	Approval policy	Guardrail (execution)	Approver identity + scope per action type (`POL-01`)	UC3
GR-05	Idempotency ledger	Guardrail (execution)	One effect per key, even under retry and replay	UC3, UC4
GR-06	Agent scopes and budgets	Guardrail (execution)	Per-agent tool allow-list; ≤ 15 calls, ≤ 60k tokens, ≤ 5 min per incident; recursion cap 2	UC2, UC3
GR-07	Disclosure filter	Guardrail (output)	A notice may mention only that customer's own orders	UC3
GR-08	No prompt override	Guardrail (input)	Event, tool and document text cannot change workflow or approval state	UC2, UC3
EV-01	Safety invariants	Eval (code)	0 unauthorised commits, 0 duplicates under replay, no negative stock, 100 % approval	UC3, UC4
EV-02	Impact accuracy	Eval (code)	Impacted order IDs and quantities = oracle	UC1
EV-03	Robustness drills	Eval (code)	Outage, timeout, duplicate and out-of-order events, restart, stale counts	UC1, UC4
EV-04	Plan and communication judge	Eval (LLM judge)	`MOD-03` ; cannot pass a case that fails `EV-01`	UC2, UC3
EV-05	End-to-end scenario pack	Eval (code + judge)	20 incidents: 5 simple, 4 conflicting data, 4 missing or duplicate events, 4 service failures, 3 approval edge cases	UC2
XP-01	Single assistant vs. governed multi-agent	Experiment	Resolution rate, safety, tokens	UC2, UC3
XP-02	Graph discovery vs. SQL joins	Experiment	Recall of affected orders, latency	UC1
XP-03	2 vs. 3 specialists	Experiment	Resolution vs. cost	UC2
OB-01	Incident timeline	Observability	Events, facts, agent recommendations, rule decisions, approvals, writes	UC1, UC3, UC4
OB-02	Action ledger	Observability	Immutable record of every proposal, approval, commit, retry and compensation	UC3, UC4
OB-03	Business dashboard	Observability	Orders at risk / mitigated, event → plan time, tokens, € and safety incidents	UC2

Pitch (5 slides)

#	Slide title (states the conclusion)	From
1	"A delay takes 26 h and 4 teams to answer; 18 % of priority orders miss the SLA"	§1
2	"Agents propose in minutes, code computes exact impact, managers approve"	§2, §4
3	"Agents can't write: only the workflow acts, once, after approval"	§3, UC3
4	"Fresh stock data is the assumption that must hold"	§5
5	"0 duplicates under fault injection, 80 % resolved: approve a 10-week pilot"	§2, §6